

Introduction to Microeconomics

Lectures 10 & 11—Consumer Theory: Preferences and Budgets

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ECON101
Fall 2024

A model of consumer Preferences

- We start with a set of objects (bundles) X . Elements from from this set, like x or y , are the choices that the consumer can make.
- Most applications consider $X \subset \mathbb{R}_+^n$.
- In this class, $X = \mathbb{R}_+^2$. A bundle x is then $x = (x_1, x_2)$. Interpret it as: x_1 is the quantity of good 1 and x_2 is the quantity of good 2.

Consumer preferences: The Questionnaire Q

Consists of a long list of questions of the following form.

$Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:

- ☐ I strictly prefer x to y ($x \succ y$).
- ☐ I strictly prefer y to x ($y \succ x$).
- ☐ I am indifferent between x and y ($x \sim y$).

A model of consumer preferences is just a long list of questions of this type and their answers!

What is a legal answer to the questionnaire?

- We need to put some restrictions (assumptions) on answers.

Restriction 1: Completeness

For any question $Q(x, y)$ exactly one of the boxes is checked. We don't allow

- Refraining to answer (no box checked).
- Checking more than one box.
- This means we exclude attitudes like:
 - They are not comparable.
 - I don't know what x is.
 - I have no opinion.
 - I prefer x over y and y over x .
- We are also not allowing attitudes like:
 - It depends on what others think.
 - It depends on the circumstances (frame).
 - Any consideration that includes **intensity**, like I somewhat prefer x , or I love x and hate y .

Restriction 2: Transitivity

Suppose I have bundles x, y, z such that $Q(x, y) : x \succ y$, and $Q(y, z) : y \succ z$. Then this implies that your answer to $Q(x, z)$ should be $x \succ z$.

Similarly, if $Q(x, y)$ is $x \sim y$ and $Q(y, z)$ is $y \sim z$ then in $Q(x, z)$ your answer should be $x \sim z$.

- Questionnaire Q could have been rephrased instead with the following type of questions:
 - $Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:
 - ☐ I consider x at least as good as y ($x \succeq y$).
 - ☐ I consider y at least as good as x ($y \succeq x$).
- What is the relationship between the answers to questionnaire Q and the one above?
 - If $x \succeq y$ and $y \succeq x$ then $x \sim y$.
 - If $x \succeq y$ but $y \not\succeq x$ then $x \succ y$.
 - If $y \succeq x$ but $x \not\succeq y$ then $y \succ x$.

Suppose the set of bundles $X \subset \mathbb{R}_+^n$ consists of *goods that are desirable*.

Restriction 3: Monotonicity or “more-is-better”

If $x \geq y$, where $\geq$ denotes the component-wise standard order, then $x \succsim y$.

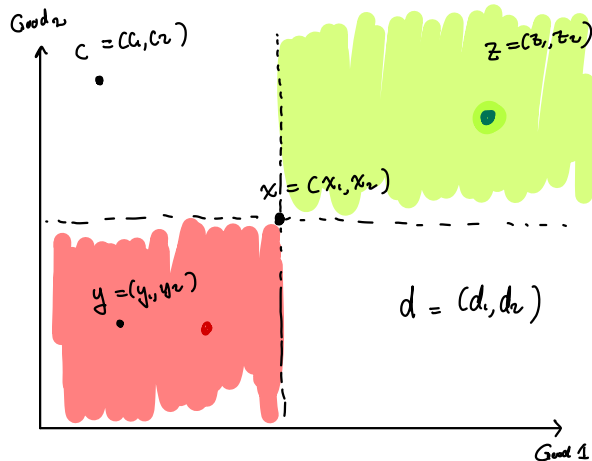
If $x > y$, where $>$ denotes the component-wise strict order, then $x \succ y$.

Example: say good 1 is food and good 2 is clothing. $x = (30, 40)$ and $y = (20, 30)$. If preferences satisfy monotonicity, then the answer to $Q(x, y)$ should be $x \succ y$.

- Any relation R over X is called a **preference relation** if it consists of a set of legal answers to the questionnaire Q , that is, if it satisfies completeness and transitivity.

Graphical representations

$X = \mathbb{R}_+^2$; Assumptions 1-3



$z \succ x$ AND $x \succ y$
 $z \succ y$

Graphical representations: Indifference curves

- For a bundle x the **indifference curve at x** is the set of all bundles that are indifferent to x :

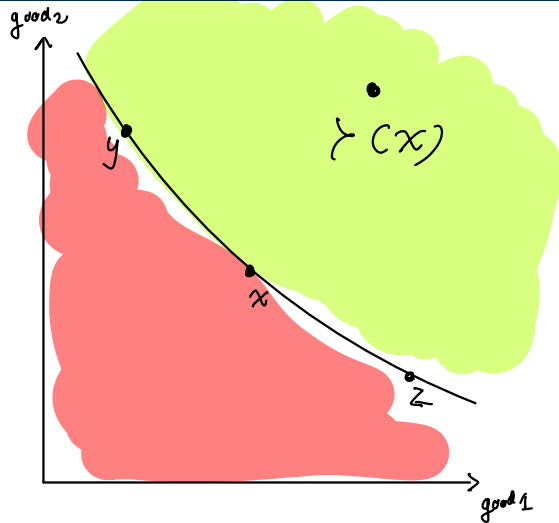
$$IC(x) = \{y \in X : y \sim x\} \quad x \sim y \sim z$$

- The strictly preferred set to x is the set of all bundles that are strictly preferred to x :

$$\succ(x) = \{y \in X : y \succ x\}$$

- The strictly worse set to x is the set of all bundles that are worse to x :

$$\prec(x) = \{y \in X : x \succ y\}$$



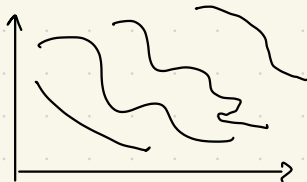
Daryn's Example at 5:12 PM

4 goods x_1, x_2, x_3, x_4

$\times (x_1, x_2)$ vs (x_3, x_4)  Not Valid

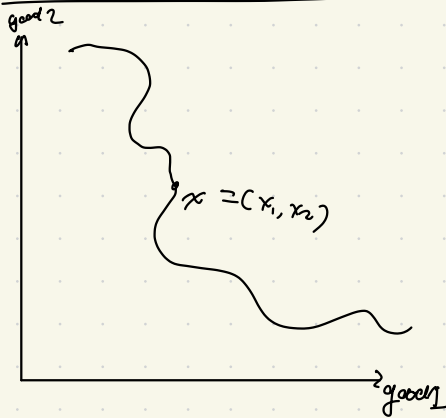
$\checkmark (x_1, x_2, 0, 0)$ vs $(0, 0, x_3, x_4)$ Is valid.

Daryn's ex. at 5:37



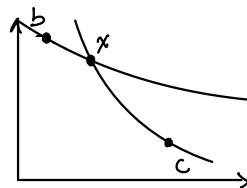
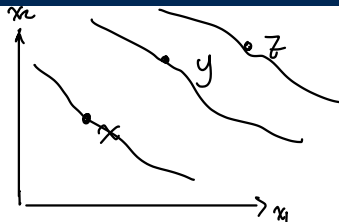
A1 and A2

Ola?'s example



Implications of the restrictions

- Completeness implies that the indifference curve map is **full**, through each bundle there is an indifference curve passing through it.
- Transitivity implies that the indifference curves **cannot intersect**!
- Monotonicity implies that the indifference curves **are not thick**!



$x \sim c$
 $b \sim x$

b vs c
• $b \succ c$
 $c \sim x, x \sim b$
 $\Rightarrow b \sim c$!
• $c \succ b$?



From Preferences to Utility

- Although preference relations provide a model of the consumers preferences, they are somewhat cumbersome to work with.
- Recall: a preference relation provides an **ordering** over the consumption bundles.
- Idea: can we get some mathematical object that is more tractable to work with, while still preserving the idea that a preference relation provides an ordering over consumption bundles?
- The answer is **Yes**!
- We say that a **utility function** $u: X \rightarrow \mathbb{R}$ represents the preference relation $\succsim$ over X if for every $x, y \in X$, $x \succsim y \Leftrightarrow u(x) \geq u(y)$.
- Given a preference relation, can we always ensure that a utility function exists?
 - If $\succsim$ is complete and transitive and X is finite or (countable), the answer is Yes.
 - On the other hand, if X is infinite, like $\mathbb{R}_+^2$, then we need an extra technical restriction: **continuity**, but this will be discussed in higher level courses.

ADVANCED
(NOT for exam)

Utility is an ordinal concept, not a cardinal one.

- Suppose that a preference relation $\succsim$ can be represented by the utility function u .
- Do the numbers $u(x)$ for each $x \in X$ matter?
- The answer is no, what matters is the order (Ranking), not the number!
- Utility is ordinal, not cardinal (the magnitude matters).
- The previous idea can be formally stated with the concept of a **monotonic transformation**:
 - Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function, i.e. if $a \geq b$ then $f(a) \geq f(b)$.
 - Let u be a utility function representing $\succsim$. Then $v(x) = f(u(x))$ represents the same preferences as u .
 - $x \succsim y \Leftrightarrow u(x) \geq u(y) \Leftrightarrow v(x) \geq v(y)$.

$$\begin{aligned} u(x_1, x_2) &= x_1 x_2 & \Leftrightarrow & x_1 x_2 \geq y_1 y_2 & \Leftrightarrow & w(x_1, x_2) = \ln(x_1) + \ln(x_2) \\ (x_1, x_2) \succsim (y_1, y_2) & \Leftrightarrow & \sqrt{x_1 x_2} \geq \sqrt{y_1 y_2} & \Leftrightarrow & \ln(y_1) + \ln(y_2) = w(y_1, y_2) \end{aligned}$$

- With the help of a utility function, it becomes easy to define an indifference curve:

$$x \sim y \Leftrightarrow u(x) = u(y)$$

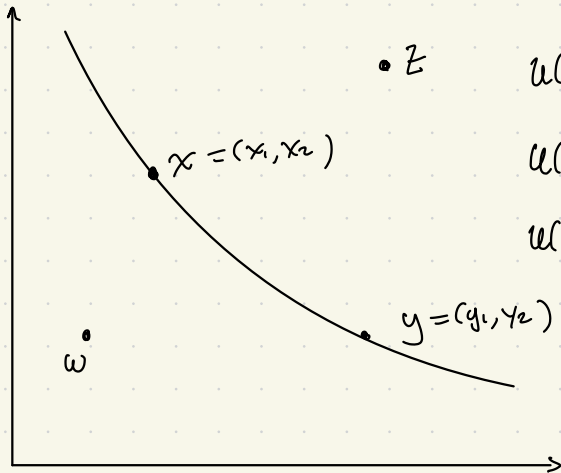
- Then the indifference curve passing through x is the set of consumption bundles that achieve the same level of utility $u(x)$:

$$IC(x) = \{y \in X : u(y) = u(x)\}$$

- We can define in a similar way the strictly preferred and the strictly worse sets.

$$P(x) = \{y \in X : u(y) > u(x)\}$$

$$L(x) = \{y \in X : u(y) < u(x)\}$$



• z

$$u(x) = u(y)$$

$$u(z) > u(x) = u(y)$$

$$u(w) < u(x) = u(y)$$

Example: $X = \mathbb{R}_+^2$

$$u(\underbrace{x_1, x_2}_{x = (x_1, x_2)}) = \sqrt{x_1 x_2}$$

↑
consumption bundle

Fix a level of utility u_0

An IC : set of bundles (x_1, x_2)
that $u_0 = \sqrt{x_1 x_2}$

↳ solve for x_2

$$u_0^2 = x_1 x_2 \Rightarrow x_2 = \frac{u_0^2}{x_1}$$

$$u_0 = 1$$

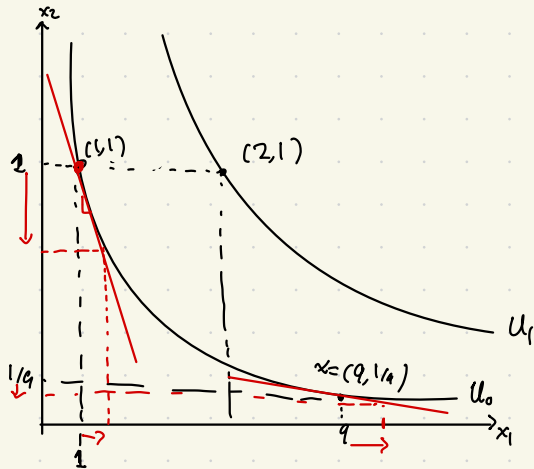
$$x_1 = 9$$

$$x_2 = 1/9$$

$$y = (2, 1)$$

$$u(y) = \sqrt{2} = u_1$$

$$\Rightarrow x_2 = \frac{2}{x_1}$$



The Marginal Rate of Substitution

- It is the absolute value of the slope of the indifference curve at a particular point.
- It measures the cost of opportunity of good 1 in terms of good 2 while staying on the same indifference curve.
- Intuitively, it is the personal opportunity cost (in terms of the consumer's preferences) of remaining in the same indifference curve, measured as units of good 2 that must be given up to get one more unit of good 1.
- How? Let's compute the total change in utility, when the utility function is $u(x_1, x_2)$:

$$\Delta U = \frac{\Delta U}{\Delta x_1} \Delta x_1 + \frac{\Delta U}{\Delta x_2} \Delta x_2$$

- At an indifference curve, utility is not changing, so $\Delta U = 0$, this leads to:

$$\frac{\Delta x_2}{\Delta x_1} = - \frac{\frac{\Delta U}{\Delta x_1}}{\frac{\Delta U}{\Delta x_2}}$$

- The **Marginal Rate of Substitution** is the absolute value of the quantity above.

$$U(x_1, x_2) \longrightarrow \Delta U = \frac{\Delta U}{\Delta x_1} \cdot \Delta x_1 + \frac{\Delta U}{\Delta x_2} \cdot \Delta x_2$$

$$\left. \begin{array}{l} \Delta U \\ = \text{small} \\ \text{change} \\ \text{in utility} \end{array} \right\} \begin{array}{l} \text{infinitesimal} \\ \text{change} \end{array} \xrightarrow{\text{differential}} dU = \frac{\partial U}{\partial x_1} \cdot dx_1 + \frac{\partial U}{\partial x_2} \cdot dx_2$$

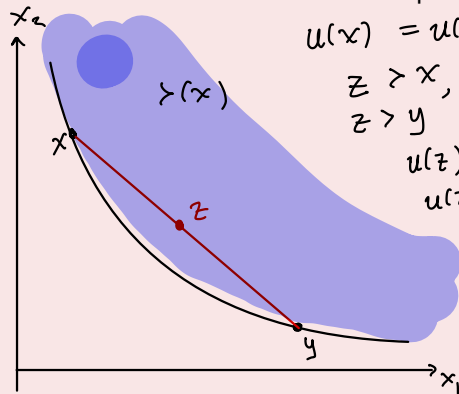
Along an indifference curve $dU = 0 \Rightarrow \frac{\partial U}{\partial x_1} \cdot dx_1 + \frac{\partial U}{\partial x_2} \cdot dx_2$

solve $\frac{dx_2}{dx_1}$

$$\frac{dx_2}{dx_1} = - \frac{\frac{\partial U}{\partial x_1}}{\frac{\partial U}{\partial x_2}} = - \frac{MU_1}{MU_2} = MRS = \text{slope of IC}$$

Convexity

Preferences that are convex exhibit a preference for more balanced consumption bundles.



$$u(x) = u(y)$$

$$z \succ x,$$

$$z \succ y$$

$$u(z) > u(x)$$

$$u(z) > u(y)$$

- IC are convex for convex preferences

- MRS is decreasing along the IC

- The strictly preferred set is a convex set.



A convex set is any set in which the line joining two points in the set is also contained in the set

Examples:

$$u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$$

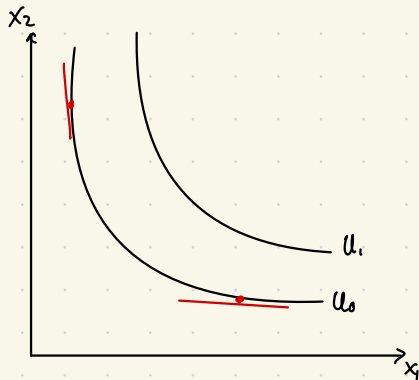
$$u_0 = \sqrt{x_1} + \sqrt{x_2}$$

$$\hookrightarrow \text{solve for } x_2 \Rightarrow x_2 = (u_0 - \sqrt{x_1})^2$$

$$MU_1 = \frac{1}{2\sqrt{x_1}}$$

$$MRS = - \frac{MU_1}{MU_2} = - \frac{\sqrt{x_2}}{\sqrt{x_1}}$$

$$MU_2 = \frac{1}{2\sqrt{x_2}}$$



Ex: Perfect Substitutes

$$U(x_1, x_2) = ax_1 + bx_2$$

$$a, b > 0$$

$$U_0 = ax_1 + bx_2$$

$$x_2 = \frac{U_0}{b} - \frac{a}{b} x_1$$

$$MRS = -\frac{a}{b} = \text{constant}$$

x_1 = coke
 x_2 = pepsi

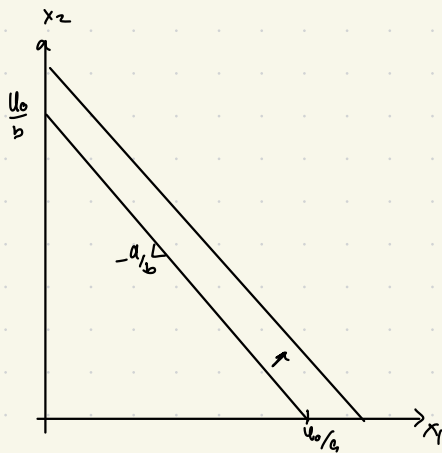
$$U(x_1, x_2) = 2x_1 + x_2$$

$$MRS = 2$$

↑ Prof. Al's preference

Arushai's Pref: $g(x_1, x_2) = x_1 + x_2$

$$\hookrightarrow h(x_1, x_2) = 2x_1 + 2x_2$$



Perfect Complement

Ex. Preferences for R & L shoes
(assuming you are not a pirate)

Ex. Espresso: Water (x_1) coffee (x_2)
2 1

$$u(x_1, x_2) = \min \left\{ \frac{x_1}{a}, \frac{x_2}{b} \right\}$$

$$a, b > 0$$

$$u(x_1, x_2) = \min \left\{ \frac{x_1}{2}, x_2 \right\}$$

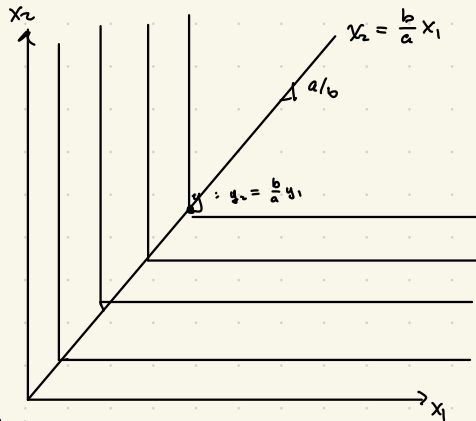
$$\begin{aligned} u(36, 18) &= u(38, 18) \\ &= u(36, 20) \end{aligned}$$

$$\textcircled{1} \quad u_0 = \min \left\{ \frac{x_1}{a}, \frac{x_2}{b} \right\}$$

$$\text{Case 1: } \frac{x_1}{a} < \frac{x_2}{b} \Rightarrow u_0 = \frac{x_1}{a} \Rightarrow a u_0 = x_1$$

$$\text{Case 2: } \frac{x_1}{a} > \frac{x_2}{b} \Rightarrow u_0 = \frac{x_2}{b} \Rightarrow b u_0 = x_2$$

$$\begin{aligned} \frac{x_1}{a} < \frac{x_2}{b} &\Rightarrow \frac{b}{a} x_1 < x_2 \\ \frac{x_1}{a} > \frac{x_2}{b} &\Rightarrow \frac{a}{b} x_1 > x_2 \end{aligned}$$



$$u(x_1, x_2) = \frac{1}{18} \min \left\{ \frac{x_1}{2}, x_2 \right\}$$

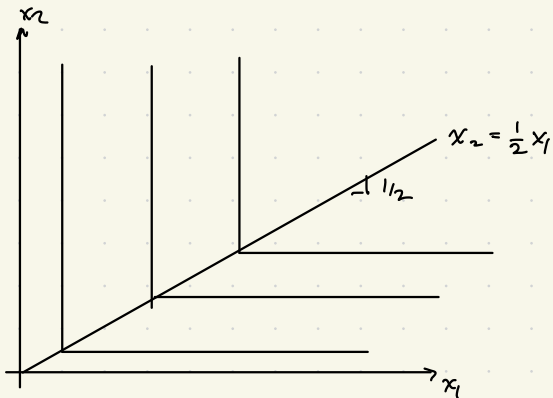
x_1 = water

x_2 = coffee

1 cup : 36 gr x_1
18 gr x_2

$$u(36, 18) = 1$$

Kink : $\frac{x_1}{2} = x_2$



The budget set and the budget line for the consumer.

- Goal: represent what is feasible to consume given market prices and income.
- Consumption space: $X = \mathbb{R}_+^n$, n goods, indexed x_i , $i = 1, \dots, n$.
- Prices $p_1, \dots, p_n$ for each good, all non-negative.
- Consumer takes prices as given, i.e., consumer cannot “bargain” for a different price.
- The budget set is the set of all **feasible** consumption bundles given prices and income I of the consumer:

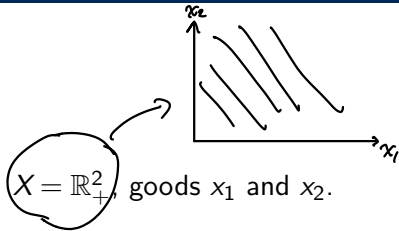
$$B(p_1, \dots, p_n, I) = \{(x_1, \dots, x_n) : p_1x_1 + p_2x_2 + \dots + p_nx_n \leq I\}$$

Total expenditures $\leq$ income.

- Intuitively: Total expenditures must be less than or equal to Income.
- When $p_1x_1 + \dots + p_nx_n = I$, this corresponds to the **budget hyperplane**, and is the set of consumption bundles that are feasible and exhaust the consumer's income.

The two-good case

- Suppose we are only considering two goods, i.e., $X = \mathbb{R}_+^2$, goods x_1 and x_2 .
- Prices are p_1 and p_2 and income I .
- The budget set is described by the simple inequality: $p_1x_1 + p_2x_2 \leq I$.
- The budget line is $p_1x_1 + p_2x_2 = I$ is the set of feasible consumption bundles which exhaust the consumer's income.



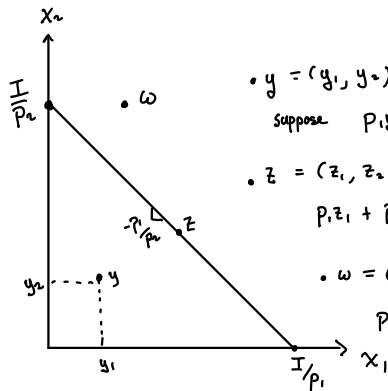
Graphical Representation

$$p_1 x_1 + p_2 x_2 \leq I \Rightarrow \text{B.L.: } p_1 x_1 + p_2 x_2 = I$$

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

$$\text{slope} = -\frac{p_1}{p_2}$$

↳ Trade off between the two goods



$$y = (y_1, y_2)$$

$$\text{suppose } p_1 y_1 + p_2 y_2 < I$$

$$z = (z_1, z_2)$$

$$p_1 z_1 + p_2 z_2 = I$$

$$w = (w_1, w_2)$$

$$p_1 w_1 + p_2 w_2 > I$$

Note

$\frac{p_2}{p_1}$: O.C. (good 2)
in terms of
good 1

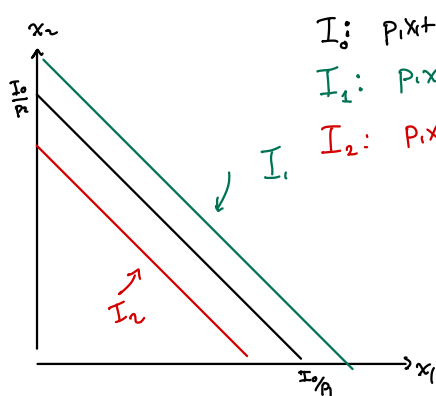
$|- \frac{p_1}{p_2}| = \frac{p_1}{p_2}$: How much it costs in the market

buying one unit of good 1 in terms of units of good 2 that the consumer needs to give up.

$$\therefore \frac{p_1}{p_2} = \text{O.C. (good 1)}$$

The slope of the budget line

Changes in Income



$$I_0: p_1 x_1 + p_2 x_2 = I_0$$

$$I_1: p_1 x_1 + p_2 x_2 = I_1$$

$$I_2: p_1 x_1 + p_2 x_2 = I_2$$

$$I_1 > I_0$$

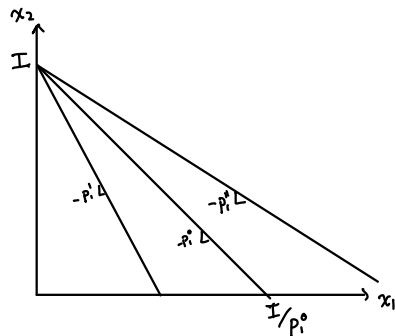
$$I_2 < I_0$$

Changes in prices

for simplicity : $p_2 = 1$ (Normalization)

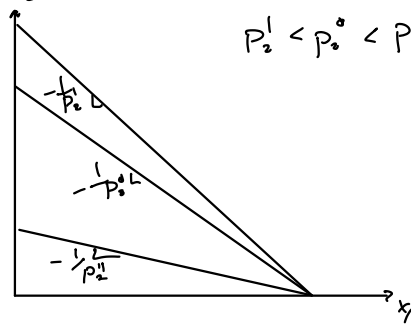
Consider changes in $p_1 \Rightarrow$ corresponds to changes in the price ratio.

$$p_1'' < p_1^o < p_1'$$



$p_1 = 1$ Changes p_2

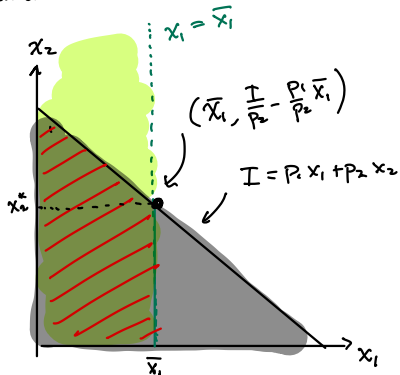
$$p_2' < p_2^o < p_2''$$



Non-linear budget constraints

Quantity Restriction: you are not allowed to consume more than $\bar{x}_1$ units of good 1: $x_1 \leq \bar{x}_1$

Consumer has income I , faces prices p_1, p_2 : $p_1 x_1 + p_2 x_2 \leq I$



 is the consumer budget set. (feasible)

How to find x_2^* ?

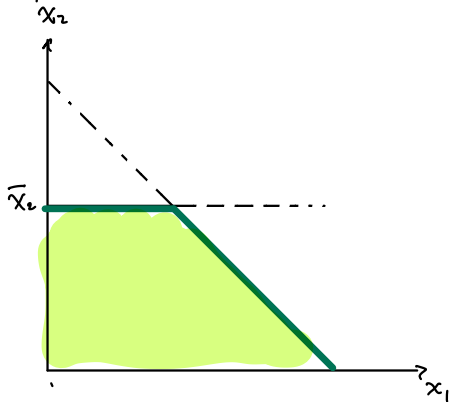
$$I = \bar{x}_1 p_1 + x_2^* p_2$$

$$\text{solve for } x_2^*: \frac{I}{p_2} - \frac{p_1}{p_2} \bar{x}_1 = x_2^*$$

$$x_2 = \begin{cases} \frac{I}{p_2} - \frac{p_1}{p_2} x_1 & \text{if } x_1 < \bar{x}_1 \\ [0, \frac{I}{p_2} - \frac{p_1}{p_2} \bar{x}_1] & \text{if } x_1 = \bar{x}_1 \end{cases}$$

Non-linear budget constraints

Quantity restriction on $x_2 = \bar{x}_2$



HW: write the analytical description.

Non-linear budget constraints

Example: Food Stamps.

Suppose $I = \$100$, $p_1 = 1$

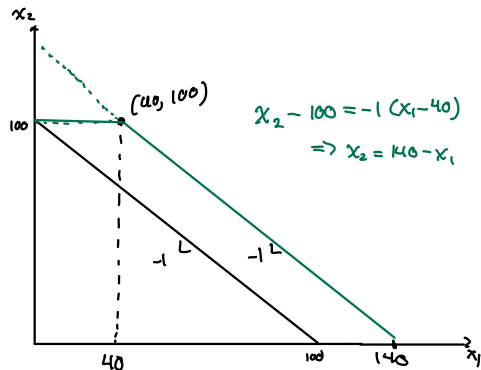
Good 1 : Food $p_2 = 1$

Good 2 : Other goods

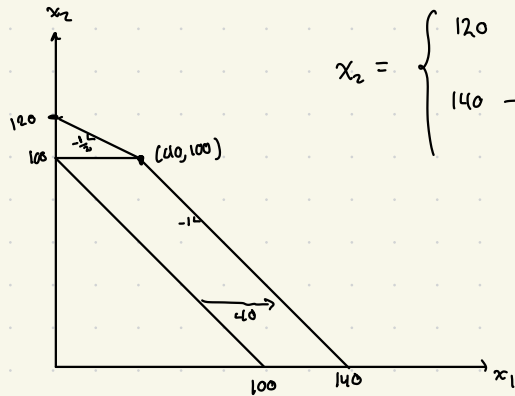
Budget Line : $x_1 + x_2 = 100$

Suppose receive \$40 worth of food stamp
What is the eq. of BL (food stamps)?

$$x_2 = \begin{cases} 100 & 0 \leq x_1 \leq 40 \\ 140 - x_1 & 40 < x_1 \leq 140 \end{cases}$$



What if there is a Black Market for Food Stamps?
 You can sell Food Stamps for \$0.50 each?



$$x_2 = \begin{cases} 120 - \frac{1}{5} x_1 & 0 \leq x_1 \leq 40 \\ 140 - x_1 & 40 \leq x_1 \leq 140 \end{cases}$$